LEVEL UP ACADEMY

Class 8 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Practice Pack: Sound

Chapter / Topic: Topic-based: use with whichever Class 8 book your students follow

For teacher use only. Do not distribute to students.

PART A: BASICS

Q1. How is sound produced?

Answer: By vibrating objects.

Q2. Which part of the body produces sound in humans?

Answer: The larynx (voice box): its vocal cords vibrate.

Q3. Can sound travel through a vacuum? Give a reason.

Answer: No. Sound needs a medium (solid, liquid or gas) to travel, and a vacuum has none.

Q4. What decides the loudness and the pitch of a sound?

Answer: Loudness depends on the amplitude of the vibration. Pitch depends on the frequency (higher frequency gives a higher pitch).

Q5. What is the audible range of the human ear?

Answer: About 20 Hz to 20,000 Hz.

PART B: THINK AND APPLY

Q6. A pendulum oscillates 40 times in 4 seconds. Find its time period and frequency.

Answer: Frequency = 40 ÷ 4 = 10 Hz. Time period = 1 ÷ 10 = 0.1 s.

Q7. A mosquito vibrates its wings 500 times per second. What is the time period of the vibration?

Answer: Frequency = 500 Hz, so the time period = 1/500 = 0.002 s.

Q8. Why does a woman's voice usually sound shriller than a man's?

Answer: Women generally have shorter vocal cords, which vibrate at a higher frequency and give a higher pitch.

Q9. Lightning and thunder happen at the same time, but we see the flash first. Why?

Answer: Light travels much faster than sound, so the flash reaches us before the sound of thunder.

Q10. What is noise pollution? Give two harmful effects and two ways to reduce it.

Answer: Unwanted, excessive sound in the surroundings. Effects (any two): lack of sleep, high blood pressure, anxiety, temporary or permanent hearing loss. Ways (any two): lower the volume of TVs and music; avoid unnecessary horns; plant trees along roads; keep noisy activities away from homes and schools.

Q11. Why do we put our ear to a railway track to hear a distant train? (Do NOT try this!)

Answer: Sound travels faster through solids (the steel track) than through air, so the train's sound reaches us sooner through the rail.